

IoT Environmental Monitor

Battery-powered sensor node with LoRaWAN uplink, e-ink display, and MQTT bridge. 6-month battery life on AAA cells.

ESP32	LoRaWAN	FreeRTOS	MQTT
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REVISION	STATUS	BATTERY	RADIO	CLASS	UPDATED
Rev B	Complete	3 × AAA	SX1262	LoRaWAN A	Sep 2026

1. Overview

A battery-powered environmental sensor node for buildings and light industrial spaces. It samples temperature, humidity, pressure, and air quality on a fixed schedule, shows the current readings on a low-power e-ink display, and uplinks over LoRaWAN to a gateway that bridges the decoded payloads onto MQTT topics.

The design target was six months on three alkaline AAA cells, and meeting it drove every other decision: e-ink rather than an LCD because it draws nothing between refreshes, LoRaWAN rather than Wi-Fi because a transmission costs milliseconds rather than seconds of radio-on time, and an aggressive deep-sleep duty cycle in which the node is awake for well under a second per fifteen-minute interval.

2. Node operation

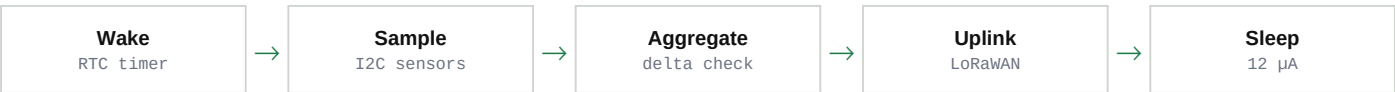


Figure 1 — A full cycle completes in roughly 640 ms; the e-ink refresh runs only when a displayed value has actually changed.

Uplinks are not sent on every cycle. The node compares each reading against the last transmitted value and skips the uplink when nothing has moved beyond its threshold, subject to a heartbeat that forces a transmission at least once an hour. In a stable room this cuts radio use by roughly two thirds, which is most of the difference between four months of battery life and six.

3. Subsystems

Power	Three AAA cells into a buck-boost holding 3.3 V to end of discharge; 12 µA deep sleep.
MCU and RTOS	ESP32-S3 with FreeRTOS; sampler, display, and radio tasks with explicit sleep gating.
Sensors	Temperature and humidity, barometric pressure, and a VOC index sensor on one I2C bus.
Radio	SX1262 driving LoRaWAN 1.0.4 Class A, OTAA join, ADR enabled, EU868 and US915 builds.
Display	2.9-inch e-ink with partial refresh for values and a full refresh once per day.
Bridge	Gateway-side decoder mapping uplinks to per-sensor MQTT topics with retained state.

4. Uplink payload

```
/* Fixed 11-byte payload, port 2. Fits SF12 with room for MAC commands. */
struct uplink_v2 {
    uint8_t  flags;           /* [0] heartbeat [1] low_batt [2] sensor_fault */
    int16_t  temp_c;         /* 0.01 C/LSB, -40.00 .. +85.00 */
    uint16_t rh_pct;         /* 0.01 %/LSB */
    uint16_t press_hpa;      /* 0.1 hPa/LSB, offset 300.0 */
    uint16_t voc_index;      /* 1..500, sensor index scale */
    uint16_t vbatt_mv;       /* measured under radio load */
} __attribute__((packed));

/* Bridge output, retained per device */
env/<dev_eui>/temperature -> { "v": 21.44, "unit": "C", "ts": 1789... }
env/<dev_eui>/humidity    -> { "v": 47.10, "unit": "%", "ts": 1789... }
env/<dev_eui>/battery     -> { "v": 3.94, "unit": "V", "ts": 1789... }
```

Listing 1 — Battery voltage is sampled during transmission, when the load is highest and the reading is most honest.

5. Power budget

STATE	CURRENT	DURATION PER CYCLE	SHARE
Deep sleep	12 µA	899.4 s	38%
Sensor sampling	4.1 mA	220 ms	6%
LoRa transmit	98 mA	120 ms (SF9)	44%
E-ink refresh	9.6 mA	300 ms, when changed	12%

Fifteen-minute cycle, one uplink in three, one display refresh in four. Projected life on 1,000 mAh alkaline cells is 6.2 months; field units have returned 5.5 to 6.8 months depending on spreading factor and temperature.

6. Field behaviour

- **Link margin.** Adaptive data rate settles most indoor nodes at SF7 to SF9. A node that cannot join falls back to a slow beacon rather than retrying continuously, which prevents a gateway outage from flattening a battery in days.
- **Cold weather.** Alkaline capacity falls sharply below 0 °C. Outdoor units use lithium primaries instead, which the firmware detects from the discharge curve and reports in the battery topic.
- **Sensor faults.** An I2C timeout marks the affected channel in the payload flags rather than blocking the cycle, so one failed sensor does not silence the node.

7. Assembly and deployment

- Enclosure is a vented ABS case with a membrane over the sensor aperture; the humidity and VOC sensors sit on a thermally isolated tab away from the regulator to keep self-heating below 0.3 °C.
- Provisioning writes the device EUI and join keys over USB during assembly and records them against the serial number, so a unit can be re-registered without opening the case.
- Roughly forty units have been in continuous service across three sites since 2025 with no hardware failures; two returns were traced to cell leakage from exhausted alkalines left in place past the low-battery flag.

8. Maintenance status

AREA	STATE	COMMITMENT
Hardware revisions	Closed	Rev B is final; design files published as-is.
Firmware	Limited	Security and LoRaWAN stack updates only.
Bridge decoder	Active	Payload decoder maintained for gateway stack changes.